

Daniel Raymond

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SUMMARY

Power electronics and embedded hardware engineer with 6+ years of PCB design, prototyping, and test. Cross-disciplinary, hands-on, and focused on reliable products.

WORK EXPERIENCE

Electrical Engineer Intern — Emvolon Inc., Woburn, MA May 2025 – Aug 2025

- Integrated PLCs/DAQs and embedded controllers; brought up and validated hardware
- Implemented requirements in schematics and test plans; supported patent-pending microgrid topology

Automotive Technician — Jay’s Garage, Mendon, MA Jun 2023 – Aug 2023

- Diagnosed and executed complex mechanical/electrical repairs; reduced turnaround and parts cost

Studio Assistant — Studio David Teeple, Northampton, MA Jul 2022 – Present

- Supported studio operations and technology integration for installations

Business Assistant — Hennessey Precision, Northampton, MA Aug 2020 – Feb 2022

- Streamlined intake/repair/billing workflows to increase shop throughput

Student Leader — FRC Team 4097 Jan 2019 – Mar 2022

- Led electronics subteam (power distribution, wiring, safety); mentored new members

PROJECTS

USB-C PD Charger Up to 240W **Class D Amplifier** 24W

- DC-input USB PD 3.1 (to 240W); GaN stage on multilayer PCB; cycle-by-cycle control and protections; 90% peak eff.
- Fully analog class-D: oscillator, feedback, filters, MOSFET drivers; ~70% eff.; validated on network analyzer

Buck Voltage Regulator ~98% **Permalux LED Lamp** Longevity

- Multiphase buck with custom switch timing for fast transients; low-loss parts; ~98% measured
- Desk lamp with replaceable CoB driver; analysis of reliability, cost, and product obsolescence

Electronic Load to 2kW **Load Switch** PWM-Controlled DC

- Modular linear and switch-mode loads for MVDC prototyping and test
- Solid-state switch for DC loads from servo PWM; designed for manufacturability and cost; sold online

EDUCATION

Worcester Polytechnic Institute (WPI) — B.S. Electrical & Computer Engineering Aug 2022 – May 2026

- GPA: 3.50/4.0 Coursework: Advanced Analog IC Design, Power Electronics, Analog Circuits, RF Design

Smith College — Dual Enrollment (High School) 2021 – 2022

- Coursework: Electronics (PHY 240)

SKILLS

Design	KiCad, Cadence Virtuoso, SolidWorks, SPICE, MATLAB, Schematics, P&IDs/PFDs
Build	SMD soldering, 3D printing, Fabrication, MIG/TIG welding
Test	LabVIEW, Oscilloscope, Network Analyzer, Power Supply, Electronic Load
Programming	Embedded C/C++, Python, Rust, HDL/Verilog
Project	GitHub workflow, Gantt charts, SOP writing, BOM management